An Explainable AI Decision Layer for Human-in-the-loop SAP ERP Workflow Integration in Supply Chain Operations

Given Name Surname
Dept. Name of Organization
City, Country
email@domain.com

Abstract

Traditional statistical process control and enterprise systems are predominantly retrospective, limiting proactive operational intervention. This study presents an SAP-oriented predictive decision framework that integrates logistics transaction data, supervised machine learning, probabilistic risk estimation, human-in-the-loop decision control, SAP-oriented workflow simulation, and cost-sensitive economic evaluation. Using 180,519 logistics observations from the DataCo Supply Chain dataset, the framework evaluates Logistic Regression, CART, Random Forest, XGBoost, and LightGBM models. Random Forest achieved the highest cross-validated ROC-AUC performance, while calibrated prediction probabilities were transformed into operational risk levels to support decision-making. The proposed decision layer converts predictive outputs into interpretable risk states, recommended actions, human review processes, workflow execution, and economic evaluation. Experimental results demonstrate that integrating predictive analytics with SAP-oriented decision processes can improve operational intervention capability and provide measurable business value. The study highlights the importance of bridging machine learning predictions with explainable and human-centered enterprise decision workflows.

Keywords

SAP ERP, IIoT Data Integration, Explainable Artificial Intelligence, Human-in-the-Loop Decision Making, Predictive Analytics, Machine Learning, Operational Optimization, Enterprise Systems

ACM Reference Format:
Given Name Surname. 2026. An Explainable AI Decision Layer for Human-in-the-loop SAP ERP Workflow Integration in Supply Chain Operations. In . ACM, New York, NY, USA, 10 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

1.1 Research Background

Enterprise Resource Planning (ERP) systems remain central to managing business processes, resource planning, and operational execution in supply chain environments. SAP-based platforms are widely used to coordinate these activities across enterprise functions [16]. At the same time, logistics and supply chain operations generate large volumes of transactional and operational data that can be used for predictive analysis and decision support [6]. Machine-learning methods provide opportunities to extract predictive information from such data and support more proactive operational decisions [20, 23]. However, integrating predictive analytics into enterprise workflows involves challenges beyond model accuracy, including interoperability, interpretability, decision transparency, and the appropriate involvement of human decision-makers [1, 15].

Conference'17, Washington, DC, USA

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

1.2 Related Research

Recent reviews of explainable AI (XAI) in decision-support systems identify interpretability, transparency, trust, and the integration of explanations into decision processes as recurring research themes [8, 9, 14]. In supply-chain applications, XAI has mainly been used to make predictive models easier to understand and to provide decision-makers with additional information about model outputs [13, 20].

A particularly relevant study by Reis et al. investigated a context-aware approach for selecting XAI methods according to the requirements of business stakeholders. The study included human participants and used supply-chain demand data to evaluate the approach [21]. This is relevant to the present work because it treats explainability as part of a decision-support setting rather than only as a property of an individual prediction model.

However, the focus of this line of research remains primarily on model explanation, stakeholder requirements, and decision support. It does not directly address the complete operational path from a predictive output to a predefined enterprise action, followed by human review when required. Similarly, the reviewed supply-chain XAI studies do not establish a specific workflow mechanism that connects risk classifications produced by a predictive model to SAP-oriented enterprise actions.

The gap considered in this study is therefore operational rather than algorithmic. The study does not propose a new XAI algorithm. Instead, it examines how predictive outputs can be organized into risk states, mapped to workflow actions, and selectively escalated to human review within an SAP-oriented supply-chain process.

1.3 Research Gap and Problem Statement

The reviewed studies show that predictive analytics, XAI, and human-centered decision support have each been investigated in supply-chain and enterprise contexts. However, these studies generally focus on individual stages of the decision process, such as prediction, model explanation, stakeholder requirements, or human interaction [9, 13, 14, 20, 21].

What remains less clearly addressed is the operational connection between these stages. In particular, a predictive result must be converted into an interpretable operational state, mapped to an appropriate enterprise action, and escalated to a human decision-maker when the case requires additional judgment. Human-AI interaction research emphasizes maintaining human understanding and control in such processes [1].

This study therefore focuses on a narrower operational gap: the organization of prediction, risk assessment, workflow action, and selective human review as one decision layer within an SAP-oriented supply-chain process. The contribution is consequently not a new prediction or XAI algorithm, but the integration of these existing capabilities into an explicit workflow-oriented decision process.

The research problem is formulated as follows:

> How can logistics and supply-chain predictions be transformed into explainable risk-based decisions, connected to SAP-oriented workflow actions, and selectively escalated for human review?

1.4 Research Objectives

The objectives of this study are:

(1) Assess the predictive performance of machine-learning models for logistics and supply-chain risk prediction.
(2) Examine whether transforming predictions into risk-based states provides more useful operational decision support.
(3) Evaluate how workflow mapping can translate risk states into predefined SAP-oriented actions.
(4) Assess the role of human review in controlling higher-risk or uncertain cases.
(5) Evaluate the overall framework in terms of predictive performance, operational effectiveness, and human-review requirements.

1.5 Research Contribution

This paper presents a workflow-oriented decision layer that uses logistics and supply-chain data to produce risk estimates and map them to predefined operational actions.

The contribution lies in combining predictive modelling, risk classification, human review, workflow simulation, and decision evaluation in a single SAP-oriented process, rather than proposing a new machine-learning or XAI algorithm.

2 Methodology

2.1 Overall Framework

The methodology follows a sequence of steps that starts with logistics data preparation and predictive modelling and then examines how model outputs can be used in an operational decision process. The study combines exploratory data analysis, supervised machine learning, probability calibration, risk classification, human review simulation, SAP-oriented workflow simulation, and cost-sensitive evaluation.

The framework consists of the following stages:

(1) data preparation and exploratory analysis;
(2) predictive model development and comparison;
(3) model diagnostics and performance analysis;
(4) construction of operational risk levels;
(5) probability calibration;
(6) cost-sensitive decision evaluation;
(7) human-in-the-loop decision simulation;
(8) SAP-oriented workflow simulation;
(9) economic evaluation and sensitivity analysis.

The sequence of the decision-layer process is shown in Fig. 1. The figure represents the workflow implemented in the study, rather than a deployed SAP system.

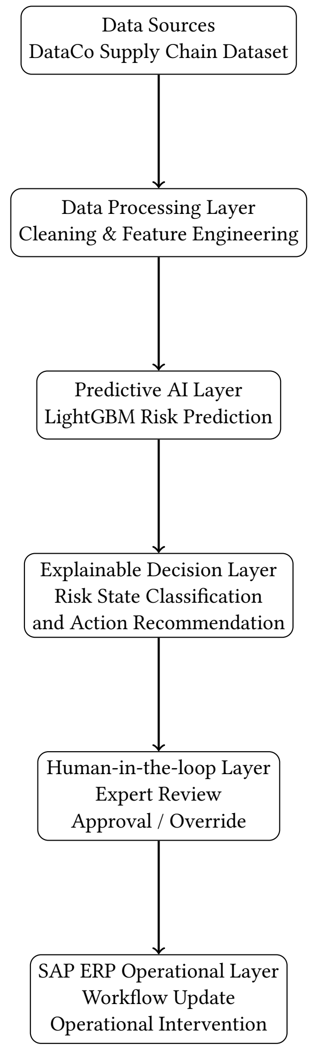

Figure 1: Decision-layer process used in the study, from prediction to risk assessment, operational action, and human review.

The predictive models estimate the probability of late delivery. These probabilities are then converted into operational risk levels and used to assign predefined actions. Cases meeting the specified review conditions are passed to a simulated human-review step. The resulting process is evaluated using predictive, operational, and economic measures.

The SAP component is treated as a workflow-oriented representation of how the resulting decisions could be connected to enterprise processes. It does not represent a live SAP implementation or direct execution of SAP transactions.

The architecture used to represent this relationship is shown in Fig. 2.

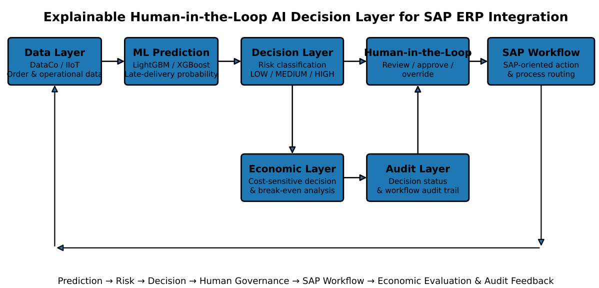

Figure 2: SAP-oriented decision-layer architecture.

2.2 Data Preparation and Process Analysis

The empirical evaluation uses the DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS dataset, originally released through Mendeley Data [6]. The dataset contains 180,519 order-level observations. The prediction target is the binary variable `Late_delivery_risk`, which indicates whether an order is classified as being at risk of late delivery.

The analysis considers operational variables related to shipping mode, market, customer segment, order region, order-item quantity, product price, and other logistics attributes available in the dataset.

The raw data are processed before model training to obtain a consistent feature representation. Categorical variables are encoded into machine-learning-compatible representations, while numerical variables are preserved unless a specific modelling procedure requires further transformation. Variables that directly reveal the target or are available only after the outcome are excluded to reduce target leakage.

The resulting feature matrix is represented as

$$X \in \mathbb{R}^{n \times p}, \tag{1}$$

where n denotes the number of observations and p denotes the number of predictive features. The corresponding target vector is

$$y \in \{0, 1\}^n. \tag{2}$$

The same processed observations are then used for the subsequent predictive modelling and decision-layer evaluation, with the specific train–test procedure described in the following subsection.

2.3 Process Capability Baseline

Process capability analysis is used as a statistical baseline to describe the historical performance of the selected logistics process. Let X denote the quality characteristic, with process mean μ, standard deviation σ, and specification limits LSL and USL. The capability index C_{pk} is defined as [17]:

$$C_{pk} = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right). \tag{3}$$

In this study, C_{pk} is used only as a descriptive baseline for historical process capability. A value below 1.33 is interpreted using the commonly applied capability criterion [17].

The capability analysis is not used to generate the prediction target or to replace the machine-learning models. Instead, it provides a process-level reference against which the subsequent transaction-level risk analysis can be interpreted.

2.4 Predictive Model Development

Five supervised learning models are evaluated for late-delivery risk: Logistic Regression, Classification and Regression Trees (CART) [3], Random Forest [2], XGBoost [5], and LightGBM [12].

For each observation i, the models estimate the probability of a late-delivery event:

$$\hat{p}_i = P(Y_i = 1 \mid X_i), \tag{4}$$

where $\hat{p}_i$ is the predicted probability and X_i is the corresponding feature vector.

Model performance is compared using cross-validation. ROC-AUC is used as the primary discrimination measure because it evaluates the ability of a classifier to distinguish positive from negative observations across decision thresholds [10].

CART is also retained as an interpretable baseline because its tree structure provides explicit decision rules. Its node impurity is measured using the Gini index [3]:

$$I(t) = 1 - \sum_{k=1}^{C} p_k^2, \tag{5}$$

where p_k denotes the proportion of observations belonging to class k at node t.

The resulting model probabilities are subsequently used as inputs to risk classification and the decision-layer analysis.

2.5 Probability Calibration

Because the decision layer uses predicted probabilities to define risk levels, calibration is evaluated in addition to discrimination [11, 19].

The Brier Score measures the accuracy of probabilistic predictions [4]:

$$BS = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2. \tag{6}$$

Expected Calibration Error (ECE) is also calculated by grouping predictions into B probability bins [11, 18]:

$$ECE = \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right|. \tag{7}$$

Here, S_b denotes the observations assigned to bin b.

These measures are used to determine whether predicted probabilities provide a sufficiently reliable basis for the subsequent risk classification and decision thresholds.

2.6 Cost-Sensitive Decision Evaluation

Predictive performance alone does not determine whether an operational intervention is preferable. The decision layer therefore evaluates alternative actions using assumed costs for late delivery and intervention. This follows the cost-sensitive decision perspective, where the consequences of classification errors are incorporated into the decision process [7].

For observation i, the expected cost of taking no intervention is defined as:

$$C_{\text{no intervention},i} = \hat{p}_i C_L, \tag{8}$$

where $\hat{p}_i$ is the predicted probability of late delivery and C_L is the assumed cost associated with a late-delivery event.

For an intervention decision, the expected cost is represented as:

$$C_{\text{intervention},i} = C_{\text{residual},i} + I_i C_I, \tag{9}$$

where $C_{\text{residual},i}$ represents the estimated remaining late-delivery cost after intervention, C_I is the assumed intervention cost, and $I_i \in \{0, 1\}$ indicates whether the intervention is applied.

The difference between the two alternatives is expressed as:

$$S = C_{\text{no intervention}} - C_{\text{intervention}}. \tag{10}$$

A positive value indicates that the intervention has a lower estimated cost under the specified assumptions. The corresponding relative difference is:

$$SR = \frac{S}{C_{\text{no intervention}}}. \tag{11}$$

These results are interpreted as scenario-based economic estimates, since the dataset does not provide directly observed intervention costs or verified financial savings. Sensitivity analysis is therefore used to examine how changes in the assumed cost parameters affect the preferred decision.

2.7 Cost Sensitivity and Break-Even Analysis

The economic evaluation depends on assumed costs for late delivery and operational intervention. A sensitivity analysis is therefore performed to examine how changes in these assumptions affect the preferred decision. This follows the cost-sensitive decision perspective, where the consequences of alternative decisions are considered alongside classification performance [7].

For each scenario, intervention is considered preferable when:

$$C_{\text{decision}} < C_{\text{baseline}}. \tag{12}$$

For a given observation, the break-even intervention cost is the difference between the expected cost without intervention and the estimated residual cost after intervention:

$$C_{I,\text{BE}} = C_{\text{baseline}} - C_{\text{residual}}. \tag{13}$$

An intervention is economically preferable when:

$$C_I < C_{I,\text{BE}}. \tag{14}$$

The analysis is used to identify how sensitive the resulting decisions are to the assumed cost parameters. Since the dataset does not contain verified intervention costs, the results are interpreted as scenario estimates rather than measured financial savings.

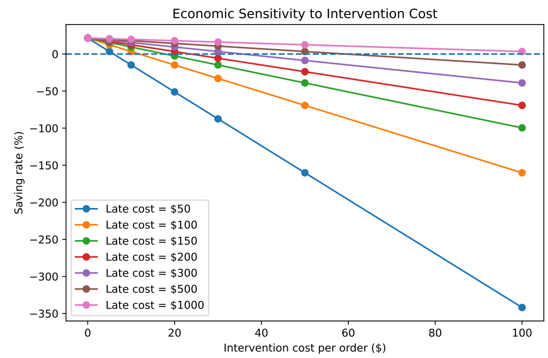

Figure 3: Sensitivity of the estimated intervention benefit to assumed late-delivery and intervention costs.

2.8 Human-in-the-Loop Decision Process

Human review is introduced as a selective control step for cases identified by the decision rules as requiring additional assessment. This follows human-AI interaction principles that emphasize user understanding, control, and the ability to intervene in AI-supported decisions [1, 22].

The evaluated decision flow is:

$$\begin{aligned} \text{Prediction} &\longrightarrow \text{Risk Assessment} \\ &\longrightarrow \text{Action Selection} \\ &\longrightarrow \text{Human Review (if required)} \\ &\longrightarrow \text{Final Decision.} \end{aligned} \tag{15}$$

LOW-risk observations follow the standard processing path. MEDIUM- and HIGH-risk observations are eligible for human review according to the predefined decision rules. Human review is simulated in the prototype and does not represent interaction with actual SAP users.

The analysis records the number and proportion of observations escalated for review. These measures are used to estimate the review workload associated with the decision policy and to assess whether the policy limits human intervention to selected cases.

2.9 SAP-Oriented Workflow Integration

The resulting decisions are represented as an SAP-oriented workflow to examine how predictive outputs can be incorporated into an enterprise decision process. This representation is a workflow simulation and does not execute SAP transactions.

The simulated workflow consists of five logical stages:

(1) logistics and operational data;
(2) machine-learning prediction;
(3) risk assessment and decision rules;
(4) human-review control;
(5) SAP-oriented workflow action.

For each observation, the decision layer produces a structured record containing the predicted late-delivery probability, assigned risk category, selected action, workflow state, human-review requirement, and final decision status.

This representation makes the transition from a probabilistic model output to an operational decision explicit. It also records the main elements of the decision path, allowing the selected action and any human intervention to be traced back to the corresponding prediction and decision rule. Traceability and human control are relevant considerations in AI-supported decision processes [1, 22].

A lightweight Streamlit prototype is used to simulate the workflow. The prototype accepts the selected operational attributes, generates the calibrated risk probability, applies the predefined decision rules, and displays the resulting workflow action and human-review status. The estimated economic effect of the selected decision is also reported.

The prototype is used for evaluation and demonstration of the decision process. It is not considered a production SAP integration, and it does not establish a connection to or execute transactions in an SAP system.

2.10 End-to-End Evaluation

The complete framework is evaluated using representative cases from the three operational risk categories: LOW, MEDIUM, and HIGH. The evaluation considers the complete decision path rather than predictive performance alone, linking the model output to the resulting risk classification, selected action, human-review status, workflow state, and estimated cost.

For each evaluated observation, the decision record is represented as:

$$D_i = \left\{\hat{p}_i, R_i, A_i, H_i, W_i, F_i, C_{\text{baseline},i}, C_{\text{decision},i}\right\}, \tag{16}$$

where $\hat{p}_i$ is the predicted late-delivery probability, R_i is the assigned risk category, A_i is the selected operational action, H_i indicates whether human review is required, W_i represents the workflow state, and F_i records the final decision status.

The end-to-end evaluation checks whether these elements remain consistent with the predefined decision rules. It also examines the number of cases requiring human review and the estimated economic effect of the resulting decisions.

This stage therefore evaluates the proposed decision layer as a complete process, from probabilistic prediction to operational decision, rather than treating model accuracy as the final evaluation criterion.

3 Results and Discussion

The results are presented in the same sequence as the evaluation procedure described in the methodology. The analysis first establishes the characteristics of the dataset and historical delivery outcomes, then evaluates predictive performance and probability calibration. The subsequent results examine risk stratification, cost-sensitive decisions, human-review workload, and the simulated SAP-oriented workflow. This structure separates model performance from the operational and economic evaluation of the resulting decisions.

3.1 Baseline Statistical Assessment

The analysis was performed on 180,519 order-level observations from the DataCo Supply Chain dataset [6]. The dataset contains operational attributes related to shipping, markets, order regions, product characteristics, and delivery performance.

The target variable, `Late_delivery_risk`, was treated as a binary outcome representing whether an order was associated with a late-delivery event.

Before predictive modelling, a baseline statistical assessment was conducted to characterize the historical data distribution and identify initial relationships between operational attributes and delivery outcomes. These analyses describe the dataset characteristics and do not represent predictive performance.

The distribution of the target variable is summarized in Table 1.

Table 1: Distribution of late-delivery risk classes.

Class	Count	Percentage
1	98977	54.830000
0	81542	45.170000

The target distribution provides an indication of the proportion of late and non-late delivery cases. This information is considered during later model evaluation because class imbalance may influence classification metrics.

Categorical associations between operational attributes and the target variable were then examined using contingency analysis and chi-square testing.

The results of these tests are presented in Table 2.

Table 2: Chi-square analysis of categorical operational attributes.

Feature	Chi_square	p_value	Degrees_of_freedom
Shipping Mode	37716.042505	0.000000	3
Market	8.650714	0.070448	4
Customer Segment	1.024740	0.599074	2
Order Region	74.933614	0.000000	22
Department Name	6.666695	0.756492	10
Category Name	42.889228	0.717981	49

The chi-square analysis identifies operational attributes that show different historical late-delivery patterns. These observations are used as descriptive evidence for feature analysis and do not imply causal relationships. Predictive relationships are evaluated separately through the machine-learning models presented in the following section.

3.2 Predictive Model Performance

Five supervised learning models were evaluated for late-delivery risk prediction: Logistic Regression, CART, Random Forest, XGBoost, and LightGBM [2, 3, 5, 12]. The models were compared using mean cross-validated ROC-AUC as the primary discrimination metric. ROC-AUC was selected because it measures the ability of a classifier to rank positive cases higher than negative cases across different classification thresholds [10].

The comparative results are presented in Table 3.

Table 3: Predictive Model Performance

analysis	model	metric	mean	std	min	max
Cross-Validation	Random Forest	ROC-AUC	0.7311	0.0026	0.7275	0.7357
Cross-Validation	CART	ROC-AUC	0.7300	0.0026	0.7251	0.7347
Cross-Validation	Logistic Regression	ROC-AUC	0.7299	0.0030	0.7264	0.7373
Gradient Boosting	XGBoost	ROC-AUC	0.7303	0.0041	NaN	NaN
Gradient Boosting	XGBoost	PR-AUC	0.8008	0.0032	NaN	NaN
Gradient Boosting	LightGBM	ROC-AUC	0.7300	0.0044	NaN	NaN
Gradient Boosting	LightGBM	PR-AUC	0.8005	0.0035	NaN	NaN
Random Seed Robustness	Logistic Regression	ROC-AUC	0.7296	0.0003	0.7291	0.7300
Random Seed Robustness	CART	ROC-AUC	0.7300	0.0004	0.7296	0.7304
Random Seed Robustness	Random Forest	ROC-AUC	0.7310	0.0003	0.7305	0.7313

Among the evaluated models, Random Forest achieved the highest mean cross-validated ROC-AUC:

$$\text{ROC-AUC} = 0.7311, \quad SD = 0.0026.$$

The remaining models showed similar performance levels, with CART, Logistic Regression, XGBoost, and LightGBM producing closely comparable ROC-AUC values. This indicates that the available operational features provide a moderate level of predictive information, while no individual model demonstrated a substantial advantage over the others.

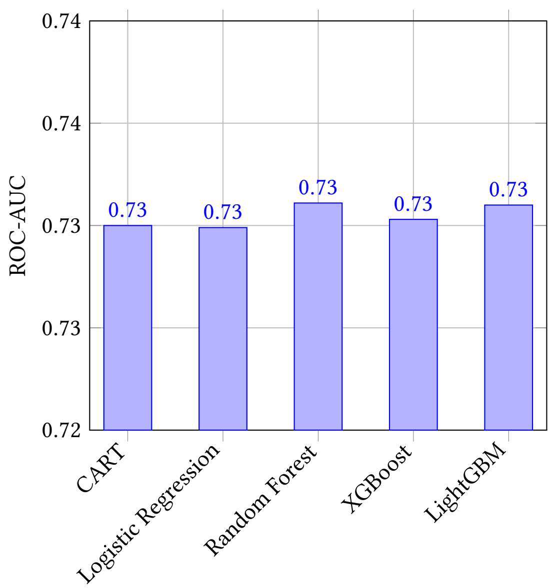

Figure 4: Comparison of predictive model performance using ROC-AUC.

Although Random Forest obtained the highest discrimination score, LightGBM was selected for the subsequent decision-layer experiments because it provides competitive predictive performance together with efficient feature analysis capabilities [12]. The objective of the following stages is not to optimize the prediction model alone, but to examine how probabilistic outputs can be transformed into operational risk categories and workflow decisions.

The classification behaviour of the selected model on the held-out test set is shown in Fig. 5.

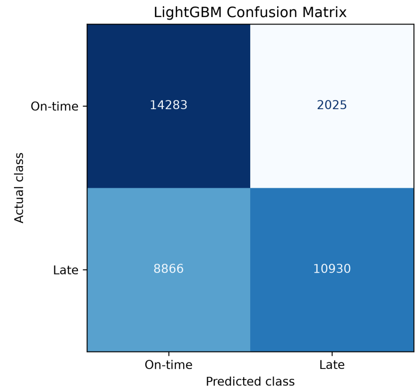

Figure 5: Confusion matrix of the selected LightGBM classifier on the held-out test set.

The feature importance analysis of the selected model is presented in Fig. 6. These values indicate the relative contribution of input variables to the model predictions and are interpreted as predictive associations rather than causal effects.

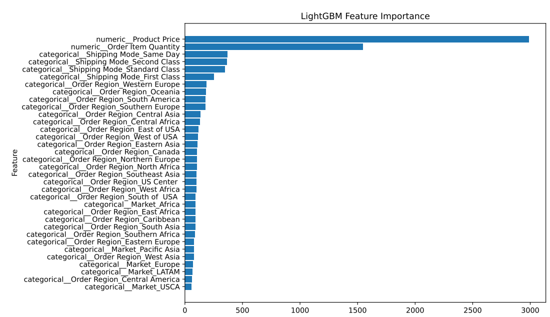

Figure 6: Feature importance analysis of the selected LightGBM classifier for late-delivery risk prediction.

3.3 Risk Stratification

The probability estimates generated by the selected predictive model were converted into three operational risk categories: `LOW`, `MEDIUM`, and `HIGH`. The purpose of this step is to transform continuous probability outputs into discrete states that can be used by the subsequent decision layer.

The risk categories were defined using predefined probability thresholds. Therefore, this analysis evaluates whether the resulting groups exhibit different historical delivery outcomes rather than optimizing the thresholds based on the test results.

The observed late-delivery rates within each risk category were:

$$\begin{aligned} P(Y = 1 \mid \texttt{LOW}) &= 0.3855, \\ P(Y = 1 \mid \texttt{MEDIUM}) &= 0.6789, \\ P(Y = 1 \mid \texttt{HIGH}) &= 0.9320. \end{aligned} \tag{17}$$

The difference between the `HIGH` and `LOW` categories was 0.5465, corresponding to an absolute difference of 54.65 percentage points in observed late-delivery rate.

The results show an increasing relationship between predicted risk category and observed late-delivery frequency. The association between risk category and delivery outcome was statistically significant ($p < 0.001$), indicating that the probability-based grouping separates historical cases according to different levels of observed delivery risk.

These results support the use of risk categories as an intermediate decision representation. However, the categories themselves do not represent automated decisions; operational actions are determined in the following decision-layer analysis, where economic assumptions and human-review requirements are considered.

Table 4: Observed late-delivery rates across risk categories.

Risk Level	Late-Delivery Rate	Operational Interpret
`LOW`	38.55%	Standard processir
`MEDIUM`	67.89%	Additional assessment ca
`HIGH`	93.20%	Priority review candi
HIGH–LOW Difference	54.65 percentage points	–

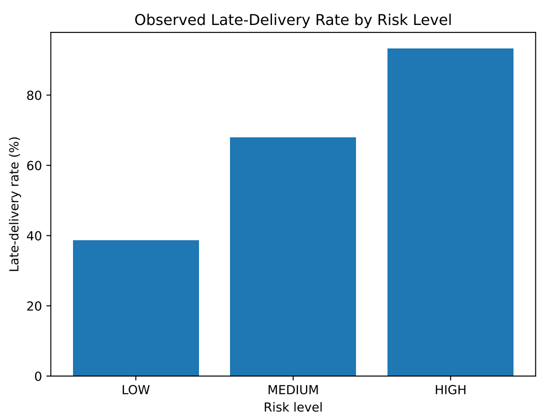

Figure 7: Observed late-delivery rates across predefined risk categories.

3.4 Probability Calibration

Since the proposed decision layer relies on predicted probabilities as risk estimates, calibration analysis was performed to examine whether the predicted probabilities correspond to the observed frequency of late-delivery events. Unlike discrimination metrics, which evaluate the ability to rank positive and negative cases, calibration evaluates the reliability of probability estimates for decision-making purposes [4, 11, 19].

Two complementary calibration measures were considered: the Brier Score and Expected Calibration Error (ECE). The Brier Score measures the mean squared difference between predicted probabilities and observed outcomes [4], while ECE measures the average difference between predicted confidence and observed event frequency across probability intervals [11, 18].

The calibration evaluation was performed on the held-out test set containing 36,104 observations. The selected LightGBM model produced the following results:

$$\mathrm{BS} = 0.1941, \qquad \mathrm{ECE} = 0.0046.$$

The obtained ECE value indicates that the average deviation between predicted probabilities and observed frequencies across calibration bins was limited. The Brier Score provides an additional measure of probabilistic prediction quality by considering both confidence and prediction error.

The calibration curve is presented in Fig. 8.

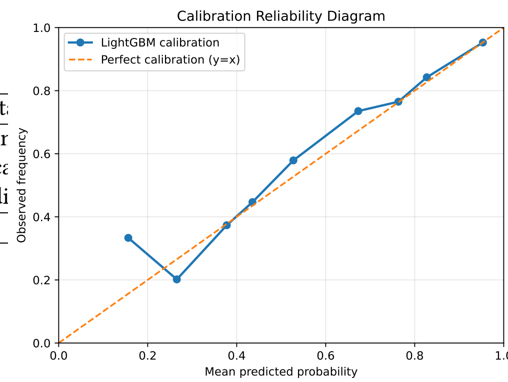

Figure 8: Probability calibration curve of the selected LightGBM model on the held-out test set.

The calibration results indicate that the model probabilities provide a reasonable representation of observed late-delivery frequencies in the evaluated dataset. This property is important for the subsequent decision-layer stages, where probability estimates are used for risk stratification and cost-sensitive evaluation.

Table 5: Probability Calibration Results

Bin	Mean Predicted Probability	Observed Frequency	Absolute Calibration Error
1	0.1564	0.3333	0.1769
2	0.2655	0.2017	0.0638
3	0.3777	0.3734	0.0043
4	0.4354	0.4467	0.0113
5	0.5274	0.5789	0.0516
6	0.6731	0.7353	0.0622
7	0.7634	0.7650	0.0017
8	0.8272	0.8424	0.0152
9	0.9532	0.9527	0.0005

3.5 SAP-Oriented Decision Layer

The calibrated probability estimates were transformed into operational risk states and mapped to SAP-oriented workflow states. The purpose of this decision layer is to represent the transition between predictive model outputs and enterprise process decisions, rather than allowing the prediction model to directly determine operational actions.

The implemented decision mapping was defined as follows:

$$\texttt{LOW} \longrightarrow \texttt{SAP\_STANDARD\_PROCESS}$$

$$\texttt{MEDIUM} \longrightarrow \texttt{SAP\_SHIPPING\_PLAN\_REVIEW}$$

$$\texttt{HIGH} \longrightarrow \texttt{SAP\_ORDER\_PRIORITY\_REVIEW}$$

These workflow states represent logical action categories used in the simulation environment. They do not correspond to executed SAP transactions; instead, they provide a structured representation of how AI-generated risk information could be associated with enterprise workflow decisions.

The decision layer therefore introduces an intermediate step between prediction and execution. It converts probabilistic outputs into interpretable operational states that can be evaluated using additional criteria, including human review requirements and economic impact.

The SAP-oriented component is implemented as a workflow simulation to demonstrate the integration pathway between predictive analytics and enterprise process logic. It is considered a proof-of-concept representation of the proposed architecture and not a production SAP deployment.

3.6 Human-in-the-Loop Decision Control

Human review was evaluated as a selective control mechanism for cases that require additional operational assessment. The purpose of this stage is not to replace the predictive model, but to define when its recommendation should be subject to human confirmation or intervention [1, 22].

Among the 36,104 held-out test observations, 12,672 were assigned to the human-review path, corresponding to a HITL rate of 35.10%.

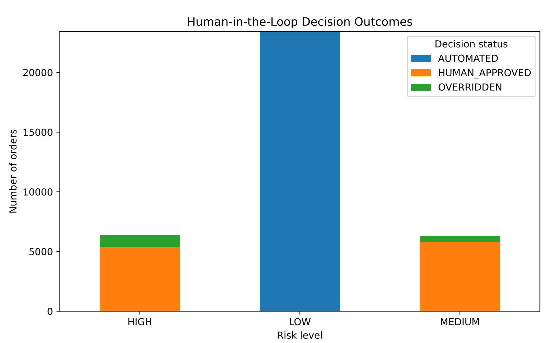

Figure 9: Simulated human-review decisions across the operational risk levels.

The result indicates that human review was required for a subset of the evaluated cases rather than for every prediction. This provides a direct measure of the review workload generated by the decision policy.

Representative `LOW`, `MEDIUM`, and `HIGH`-risk cases were also passed through the simulated workflow to verify that risk assessment, action selection, human review, and final decision states were applied consistently. These tests validate the implemented workflow logic but do not constitute evidence of performance with actual operational users.

3.7 Economic Impact and Sensitivity Analysis

The economic analysis evaluates whether the simulated decision policy provides a lower expected cost than the baseline policy under the assumed late-delivery and intervention costs. The analysis is scenario-based and does not represent observed financial savings.

Under the primary cost configuration, the estimated baseline cost was 9,897,690.45, while the estimated cost after applying the decision policy was 9,561,435.12. The resulting estimated saving was 336,255.34, corresponding to a saving rate of 3.40%.

Table 6: Scenario-based economic evaluation of the decision policy.

Metric	Value
Baseline expected cost	9,897,690.45
Decision-policy expected cost	9,561,435.12
Estimated saving	336,255.34
Estimated saving rate	3.40%
Beneficial scenarios	30/49

The result indicates a lower estimated cost under the primary scenario, but the magnitude of the benefit depends on the assumed cost parameters. Therefore, a sensitivity analysis was performed using 49 combinations of late-delivery and intervention costs. The decision policy produced a positive estimated saving in 30 of the 49 evaluated scenarios.

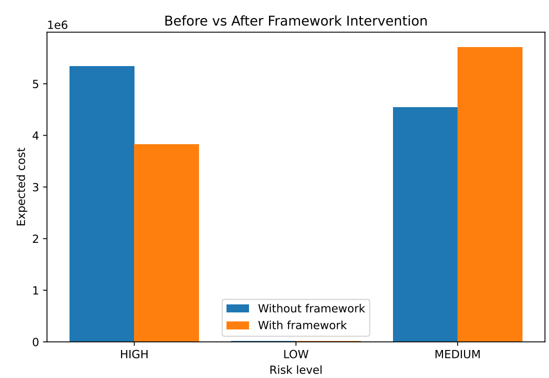

Figure 10: Estimated baseline and decision-policy costs under the primary cost scenario.

The sensitivity results therefore provide a more conservative interpretation of the economic effect: the decision policy is not assumed to be beneficial under all cost conditions, but its economic effect is examined across alternative assumptions.

3.8 Break-Even Analysis

Break-even analysis was used to estimate the intervention-cost boundary under which the decision policy remains economically preferable to the baseline. The analysis considers the relationship between the assumed cost of a late-delivery event and the maximum intervention cost that can be tolerated before the estimated benefit becomes zero.

Seven late-delivery cost levels were evaluated, ranging from $50 to $1,000. Across the evaluated scenarios, the estimated break-even intervention cost increased as the assumed late-delivery cost increased.

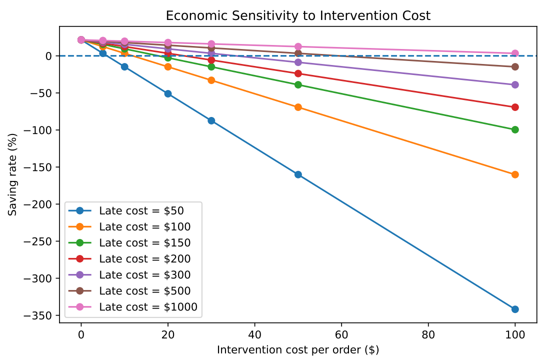

Figure 11: Estimated break-even intervention cost under different late-delivery cost assumptions.

The result indicates that the economic feasibility of intervention is sensitive to the relative magnitude of the potential late-delivery loss and the intervention cost. These values are scenario-based estimates and should therefore be interpreted as decision thresholds rather than observed financial outcomes.

3.9 SAP-Oriented Workflow Validation

The decision layer was evaluated using representative `LOW`, `MEDIUM`, and `HIGH`-risk transaction cases. The test cases were passed through the simulated workflow to verify the consistent application of risk classification, action selection, human review requirements, and final decision states.

The generated decision records retained the main attributes produced during processing, including the predicted probability, risk category, selected action, workflow state, and human-review status. This provides a traceable representation of the decision path within the simulation and supports reproducibility of the evaluated workflow logic.

The validation concerns the implemented workflow simulation and should not be interpreted as validation of a production SAP integration or of actual organizational governance processes.

4 Discussion

The results provide evidence for a structured transition from transaction-level prediction to operational decision support. The predictive models achieved moderate discrimination, with relatively small differences between the evaluated classifiers. This suggests that model selection alone does not determine the practical value of the framework.

The subsequent analyses addressed the stages that follow prediction. Calibrated probabilities were used to construct three risk categories, and the observed late-delivery rates increased from the LOW to the HIGH category. This provides empirical support for using the model output as an input to risk-based decision rules rather than treating the prediction as a final decision.

The economic analysis further showed that the usefulness of an intervention depends on its assumed costs. The primary scenario produced an estimated saving, while the sensitivity analysis showed that positive benefit was not obtained under all evaluated cost configurations. The break-even analysis therefore provides an additional decision boundary for determining when an intervention may no longer be economically justified.

Human review was implemented as a selective control step. The simulation generated a measurable review workload rather than assuming that every prediction requires human intervention. The SAP-oriented workflow simulation then represented how the resulting risk state, action, review status, and final decision could be organized within an enterprise workflow.

Taken together, these results address the research problem at the level supported by the study: they demonstrate a simulated pathway for transforming logistics predictions into risk-based decisions, economic assessments, human-review states, and SAP-oriented workflow records. The study does not establish the effectiveness of a production SAP integration or of human decision-making with real operational users.

References

[1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jenna Suh, Shamsi Iqbal, John Bennett, Kori Inkpen, Jaime Teevan, Patrice Kikin-Gil, and Eric Horvitz. 2019. Guidelines for human-AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), 1–13. doi:10.1145/3290605.3300233

[2] Leo Breiman. 2001. Random forests. *Machine Learning* 45, 1 (2001), 5–32. doi:10.1023/A:1010933404324

[3] Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. 1984. *Classification and Regression Trees.* Wadsworth, Belmont, CA.

[4] Glenn W. Brier. 1950. Verification of forecasts expressed in terms of probability. *Monthly Weather Review* 78, 1 (1950), 1–3. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[5] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.* ACM, San Francisco, CA, USA, 785–794. doi:10.1145/2939672.2939785

[6] Fabian Constante, Fernando Silva, and António Pereira. 2019. *DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS.* doi:10.17632/8gx2fvg2k6.5

[7] Charles Elkan. 2001. The foundations of cost-sensitive learning. In *Proceedings of the 17th International Joint Conference on Artificial Intelligence*, Vol. 2. 973–978.

[8] Ijeoma Noella Ezeji, Matthew Adigun, and Olukayode Oki. 2024. Computational complexity in explainable decision support system: A review. *Journal of Intelligent and Fuzzy Systems* (2024). doi:10.3233/JIFS-219407

[9] Marco Fanfani, Luciano Alessandro Ipsaro Palesi, and Paolo Nesi. 2026. Human-Centered AI for Decision Support Systems: A Systematic Review of Application Domains, Architecture Designs, Current Trends and Future Directions. *Big Data and Cognitive Computing* 10, 6 (2026), 186. doi:10.3390/bdcc10060186

[10] Tom Fawcett. 2006. An introduction to ROC analysis. *Pattern Recognition Letters* 27, 8 (2006), 861–874. doi:10.1016/j.patrec.2005.10.010

[11] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning.* 1321–1330.

[12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems 30.* Curran Associates, Inc., Long Beach, CA, USA, 3146–3157.

[13] Edward Kosasih, Emmanuel Papadakis, George Baryannis, and Alexandra Brintrup. 2024. A review of explainable artificial intelligence in supply chain management using neurosymbolic approaches. *International Journal of Production Research* 62, 4 (2024), 1510–1540. doi:10.1080/00207543.2023.2281663

[14] Georgios Kostopoulos, Gregory Davrazos, and Sotiris Kotsiantis. 2024. Explainable Artificial Intelligence-Based Decision Support Systems: A Recent Review.

Electronics 13, 14 (2024), 2842. doi:10.3390/electronics13142842

[15] Xin Li et al. 2024. Explainable artificial intelligence in supply chain management: A comprehensive review and future directions. *International Journal of Production Economics* 268 (2024), 109123.

[16] Ellen Monk and Bret Wagner. 2006. *Concepts in Enterprise Resource Planning* (2nd ed.). Thomson Course Technology.

[17] Douglas C. Montgomery. 2019. *Introduction to Statistical Quality Control* (8th ed.). Wiley, Hoboken, NJ.

[18] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. 2015. Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 29. doi:10.1609/aaai.v29i1.9602

[19] Alexandru Niculescu-Mizil and Rich Caruana. 2005. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning*. 625–632. doi:10.1145/1102351.1102430

[20] Femi Olan, Konstantina Spanaki, Wasim Ahmed, and Guoqing Zhao. 2024. Enabling explainable artificial intelligence capabilities in supply chain decision support making. *Production Planning & Control* 35, 8 (2024), 808–819. doi:10.1080/09537287.2024.2313514

[21] Marcelo I. Reis, João N. C. Gonçalves, Paulo Cortez, M. Sameiro Carvalho, and João M. Fernandes. 2025. A context-aware decision support system for selecting explainable artificial intelligence methods in business organizations. *Computers in Industry* 165 (2025), 104233. doi:10.1016/j.compind.2024.104233

[22] Ben Shneiderman. 2020. Human-centered artificial intelligence: Reliable, safe and trustworthy. *International Journal of Human–Computer Interaction* 36, 6 (2020), 495–504. doi:10.1080/10447318.2020.1741118

[23] Xun Xu, Yuqian Lu, Birgit Vogel-Heuser, and Lihui Wang. 2021. Industry 4.0 and Industry 5.0—Inception, Conception and Perception. *Journal of Manufacturing Systems* 61 (2021), 530–535. doi:10.1016/j.jmsy.2021.10.006